Prasad Sadanand Mahabare

prasadmahabare8@gmail.com | +918484976195 | linkedin.com/prasad-mahabare | github.com/MahabarePS

TECHNICAL SKILLS

- **Networking: TCP/IP, DNS, DHCP, SSH, VPN, Routing & Switching, OSI Model**
- **Monitoring & Observability: Cisco Prime, LibreNMS, Grafana**
- **Infrastructure & Network Management: Cisco ISE, Infoblox**
- **Automation: Python, Bash, Ansible**
- **DevOps: Git, GitHub, Docker**
- **Cloud: AWS (EC2, S3), Azure**

PROFESSIONAL EXPERIENCE

Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Designed and implemented **Python automation scripts** to streamline firmware upgrades across **800+ enterprise network devices**, improving deployment reliability through **Ansible** while reducing manual effort and supporting **system administration** and **infrastructure automation**.
- Collaborated with cross-functional teams to maintain Purdue University's cloud infrastructure, performing **network monitoring**, **troubleshooting**, and **incident management** to improve system reliability, availability, and user experience.
- Configured and deployed **Cisco Catalyst switches (2960S–9300 series)**, **DNS, and VPN** solutions while supporting **enterprise infrastructure**, **network security**, and **root cause analysis** of connectivity and performance issues.

PROJECTS

ESPForge - Production OTA Firmware Management Platform | Github

- Developed **WiFi-enabled ESP32** software for remote **OTA upgrades**, **watchdog-based recovery**, and live device telemetry.
- Integrated **FreeRTOS** task orchestration with **MQTT** messaging for deployment pipelines and remote maintenance.
- Developed a cloud-connected dashboard for monitoring firmware versions, OTA deployment status, device telemetry, and node health.

Enterprise Network Automation using Python, Ansible & SNMPv3

- Developed **Python automation scripts** to securely migrate Cisco IOS devices to **SNMPv3**, leveraging GitLab CI pipelines and HTTPS deployment workflows to improve **network monitoring**, **configuration management**, and operational efficiency.
- Performed **code reviews**, **debugging**, **testing**, and **log analysis** across an enterprise environment of over **800 network devices**, identifying issues through **troubleshooting** and **root cause analysis** to ensure reliable software deployments.
- Developed **Ansible playbooks** to automate Cisco IOS XE upgrades, validate deployment status, generate deployment logs, and improve **system monitoring**, **automation**, and infrastructure reliability within enterprise test environments.

EDUCATION

Master of Science in Computer Science **Jan 2022 - Dec 2023**
Purdue University, Fort Wayne, Indiana
Bachelor of Engineering in Computer Engineering **Aug 2016 - Aug 2019**
Pune University, Pune, India